CS768 Midsem Report
Neural Tree-Based Boosted Graph Learning

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Abstract

Graph Neural Networks (GNNs) have demonstrated strong performance on relational data, while Gradient Boosted Decision Trees (GBDTs) remain highly effective for heterogeneous tabular features. The Boosted Graph Neural Network (BGNN) framework integrates these paradigms by iteratively refining node representations through boosting followed by graph convolution. However, the original BGNN relies on non-differentiable decision trees and limited GNN architectures, restricting optimization flexibility and representational capacity.

This project proposes a redesigned hybrid architecture that replaces classical GBDT modules with differentiable Neural Decision Trees and substitutes standard GCN/GAT backbones with more expressive alternatives such as GraphSAGE, Graph Isomorphism Networks (GIN), and Transformer-based GNNs. The objective is to investigate whether fully differentiable boosting combined with modern graph encoders improves convergence stability, representation learning quality, and predictive performance on benchmark graph datasets.

Introduction

Graph-structured data appears in domains such as citation networks, recommendation systems, biological interaction networks, and social platforms. Learning effective representations from such data requires models that capture both node attributes and relational structure. Graph Neural Networks (GNNs) address this challenge by propagating information across neighboring nodes using message-passing mechanisms. Architectures such as Graph Convolutional Networks (GCNs) and Graph Attention Networks (GATs) have shown strong performance in node-level prediction tasks.

Despite their success, GNNs often struggle when node features are heterogeneous or tabular in nature. In contrast, Gradient Boosted Decision Trees (GBDTs) are highly effective for structured data due to their ability to model complex feature interactions and non-linear relationships. The BGNN framework combines these complementary strengths by iteratively refining node features using boosting before applying graph convolution.

However, the original BGNN framework has two key limitations. First, it relies on non-differentiable decision trees, preventing end-to-end gradient optimization. Second, it primarily evaluates standard GNN architectures, limiting exploration of more expressive graph encoders. These limitations motivate the proposed redesign.

Problem Formulation

Let a graph be defined as:

$$G = (V, E)$$

where V denotes the set of nodes and E denotes the set of edges. Each node $v_i \in V$ has a feature vector $x_i \in R^d$ and label y_i.

The original BGNN model learns a function:

$$f(X, G) = \text{GNN}(\text{GBDT}(X), G)$$

In this project, we propose the modified formulation:

$$f(X, G) = \text{GNN}_{\theta_g}\big(\text{NeuralTree}_{\theta_t}(X), G\big)$$

where:

- $\text{NeuralTree}_{\theta_t}$ is a differentiable tree-based learner,
- GNN_{θ_g} is selected from advanced architectures such as GraphSAGE, GIN, or Transformer-based GNN.

The objectives are:

1. Replace non-differentiable boosting with differentiable neural trees.
2. Compare multiple GNN backbones for representation learning.
3. Evaluate convergence behavior, accuracy, and stability.
4. Analyze the impact of feature refinement on graph prediction tasks.

Literature Review

Graph Neural Networks

Graph Neural Networks generalize deep learning to relational data through neighborhood aggregation. GCN introduced spectral graph convolution, while GAT incorporated attention mechanisms to weigh neighbor contributions. GraphSAGE enabled inductive learning by sampling neighborhoods and learning aggregation functions. GIN improved expressiveness by matching the discriminative power of the Weisfeiler–Lehman graph isomorphism test. More recently, Transformer-based GNNs utilize global attention to capture long-range dependencies.

Gradient Boosted Decision Trees

GBDT is an ensemble learning method that sequentially fits decision trees to residual errors. It is widely used for tabular data due to its robustness and ability to model non-linear feature interactions. However, classical decision trees are non-differentiable and cannot be jointly optimized with neural networks using gradient descent.

Boosted Graph Neural Networks

BGNN integrates boosting and graph learning by iteratively refining node features using decision trees before applying graph convolution. This hybrid approach improves performance when both feature quality and graph structure are important. However, the use of classical trees restricts end-to-end optimization.

Neural Decision Trees

Neural decision trees introduce soft decision boundaries, replacing hard splits with differentiable routing mechanisms. This allows gradient-based optimization and seamless integration with deep learning architectures. Such models combine the interpretability of trees with the flexibility of neural networks.

Research Gap

Existing hybrid graph-boosting frameworks rely on non-differentiable trees and limited GNN architectures. A fully differentiable boosting module combined with modern graph neural architectures remains underexplored. This project aims to address this gap through architectural redesign and empirical evaluation.

Experimental Setup

The original GBDT module in BGNN was replaced with differentiable Neural Decision Trees using soft routing mechanisms. This enables end-to-end gradient-based optimization.

To evaluate the effectiveness of the proposed redesigned BGNN architecture, experiments were conducted on two heterogeneous node classification datasets:

- **House_class**
- **VK_class**

These datasets were originally introduced in the BGNN paper for evaluating hybrid boosting-graph architectures on heterogeneous graph data with tabular node features.

Architectures Evaluated

The original BGNN framework primarily utilized GAT-based graph learning. In this project, we extend the framework by replacing the graph encoder with more expressive GNN architectures:

- **GraphSAGE**
- **Graph Isomorphism Network (GIN)**
- **Transformer-based GNN**

All experiments were evaluated across five random seeds, and final results are reported using mean accuracy and standard deviation.

Evaluation Metric

Since both datasets correspond to node classification tasks, performance is evaluated using classification accuracy.

Experimental Results

Table 1 compares the proposed architectures with the original BGNN and Res-GNN results reported in prior work.

Table 1: Comparison of BGNN Variants on Classification Datasets

2*Method	House_class		VK_class	
	Accuracy	Gap (%)	Accuracy	Gap (%)
	Original BGNN Paper			
Res-GNN	0.625 ± 0.01	-0.06	0.603 ± 0.00	4.45
BGNN	0.682 ± 0.00	9.18	0.683 ± 0.00	18.30
	Our Proposed Models			
NDT-BGNN-SAGE	0.756 ± 0.013	20.96	0.850 ± 0.005	47.31
NDT-BGNN-GIN	0.748 ± 0.011	19.68	0.842 ± 0.005	45.93
NDT-BGNN-Transformer	**0.759 ± 0.013**	**21.44**	**0.856 ± 0.006**	**48.35**

Gap Calculation

Following the original BGNN paper, relative improvement is computed with respect to the GAT baseline:

$$\text{Gap (\%)} = \frac{\text{Accuracy}_{model} - \text{Accuracy}_{GAT}}{\text{Accuracy}_{GAT}} \times 100$$

where Accuracy_{GAT} corresponds to the baseline GAT performance reported in the literature.

Analysis

The experimental results demonstrate that replacing the original GAT backbone with more expressive graph neural architectures significantly improves classification performance.

House_class

For the House_class dataset:

- BGNN-SAGE improves accuracy from 68.2% to 75.6%.
- BGNN-GIN achieves 74.8% accuracy.
- Transformer-based BGNN achieves the best performance with 75.9% accuracy.

The results suggest that advanced neighborhood aggregation mechanisms improve representation learning on heterogeneous housing graphs.

VK_class

For the VK_class dataset:

- BGNN-SAGE achieves 85.0% accuracy.
- BGNN-GIN achieves 84.2% accuracy.
- Transformer-based BGNN achieves the best performance with 85.6% accuracy.

The substantial improvement over the original BGNN baseline indicates that Transformer-based message passing effectively captures long-range dependencies and complex feature interactions in large social-network graphs.

Observations

The experiments reveal several important insights:

1. All proposed architectures outperform the original BGNN framework.
2. Transformer-based BGNN consistently achieves the best overall performance.
3. GraphSAGE provides strong and stable improvements through efficient neighborhood aggregation.
4. GIN demonstrates competitive performance while maintaining expressive message passing capability.

Connection with Original BGNN Paper

The original BGNN paper demonstrated that combining boosting with graph learning significantly improves performance on heterogeneous tabular graph datasets.

This project extends the original framework by exploring stronger graph encoders beyond standard GAT/GCN architectures.

The experimental results validate the hypothesis that modern graph neural architectures can further improve hybrid boosting-graph learning systems.

Conclusion

This project extends the BGNN framework using advanced graph neural architectures including GraphSAGE, GIN, and Transformer-based GNNs.

Experimental evaluation on House_class and VK_class demonstrates substantial improvements over the original BGNN architecture, with Transformer-based BGNN achieving the strongest performance.

The results indicate that combining boosting-based feature refinement with expressive graph representation learning is a promising direction for heterogeneous graph machine learning.

Future work includes:

- Optimization and scalability of the neural decision tree module

- Fully end-to-end optimization
- Evaluation on larger benchmark datasets
- Scalability and convergence analysis

Github link

https://github.com/user-Saranya/bgnn